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Problem Chosen

A**2006 Mathematical Contest in Modeling (MCM) Summary Sheet**

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

This paper is aimed to design an optimal schedule for the irrigation system for the given field. After estimating the spacing between sprinklers and the number of sprinklers, we propose a model to seek the minimum time required to maintain this irrigation system.

By maximizing the value of Coefficient Uniformity, we prove the conclusion: the water can be applied as uniformly as possible when the spacing between sprinklers equals to the spray radius of the sprinkler. Thus, the spacing and number of sprinklers can be determined. In order to design the optimal schedule, two principles are proposed: Principle of Optimal Positioning and Principle of Minimum Times of Moving. Based on the principles, "Heuristic Algorithm" is adopted, with a goal to find the minimum times of moving. By using our algorithm, when the spray radius is 5 meters, the spacing between sprinklers is 5 meters, the number of sprinklers is 5, and the minimum time of irrigating the whole field per time is 3.459 hours. Then we design the irrigation schedule to move the pipe, including where to move.

For the sake of the practicality, we propose three other models: Water Balance Model is employed to calculate the water requirements once; Irrigation Quota Model is proposed to determine the Irrigation Period for one season; Sub-Pipes Model is designed in order to reduce water requirement and the expense of the system.

Optimal Design of Irrigation Schedule

电子科技大学

胡方靖2302002003

张宏涛2306001016

冯涛2301308025

1. Introduction● **Background**

Sprinkler irrigation has been an important part of agricultural production in the western United States since the 1950s. About 48 percent (Irrigation Journal, 2000) of the irrigated acres in the 17 western states are watered with sprinklers, including hand-move, wheel-move, center pivot and other types. Sprinklers can be good investments when properly designed, installed, maintained and managed. A well-designed sprinkler system applies water uniformly to the soil surface and is capable of applying enough water to meet the peak demands of the crop without producing excess runoff.

In this paper, we focus on “hand move” sprinkler systems. In a typical hand move sprinkler irrigation system a mainline is installed from the pump to the field where a pipe breaks the field into small blocks. The pipe has a series of hydrants with take-off valves to provide water to the lateral. The pipe and laterals are aluminium pipes with couplers and fittings made of aluminium or galvanized steel. The couplers and fittings (because of the gaskets) allow the pipes to be easily drained, un-coupled, moved and re-coupled. This type of irrigation system is used here to irrigate the field that is 80 meters by 30 meters.

● Analysis of Problem

As mentioned above, this type of irrigation system is cheaper and easier to maintain than other systems. However, there exists a disadvantage that it requires a great deal of time and effort to move and set up the equipment at regular intervals. Therefore, our goals are:

- Minimize the amount of time required by a rancher to maintain this system;
- Water should be applied in the field as uniformly as possible.

We should determine:

- The number of sprinklers and the spacing between sprinklers;
- An optimal schedule of positioning and moving sprinkler system.

To be brief, we can regard the spray drainage of a sprinkler as an approximate circle. Because the amount of water decreases from its center to the edge, thus the water cannot be applied as uniformly as possible. Therefore, Uniformity Coefficient (Christiansen, 1942) would be adopted to measure the uniformity of water that applied to the field. By maximizing the Uniformity Coefficient, we seek to find the relation between numbers of sprinklers and spray radius. Accordingly, the numbers of the sprinklers and the spacing between sprinklers can be determined.

Then we consider that the amount of time required to maintain the system is composed of two parts: the time of positioning the pipe and the time spending on equipments, such as un-coupling, moving and re-coupling the pipe. Based on the assumption that the time of un-coupling and re-coupling the pipe is short, and out of our consideration, the first goal can be changed into minimizing the total times that required for a rancher to move the pipe and irrigate the given field. Based on these, we try to propose an algorithm to find the minimal times

2. Assumptions

- The wind and the temperature don't influence the sprinkler system;
- The irrigated field is flat, we don't take terrain condition into account;
- Head loss is ignored and the flow rate remains the same from the pipe to all the nozzles;
- The pipe can be extended via a tube that connected between the mainline from water source and the pipe, so as to enable the pipe be moved in the field;
- The time of un-coupling and re-coupling the pipe is short, and out of our consideration.

3. Symbols and Terms

Variables	Definition	Units
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C_u	Uniformity Coefficient, which measures the applied water's uniformity	/
ρ	The Precipitation Rate (Pr). It describes the volume of water sprayed from the nozzle	m/h
n	The numbers of sprinklers.	/
R	The spray range of the nozzle	m
η	Application Rate, application efficiency varies with field conditions, 80 percent is a reasonable value	/
n_i	Irrigation times.	/
t	Regular intervals	<i>minutes</i>
T_m	The amount of time required by a rancher to maintain the irrigation system.	<i>hours</i>
T	The amount of time required by a rancher to maintain the irrigation system. $T = t \times T_m$	<i>hours</i>
a	Cover rate. It defines the percentage that the irrigation system can irrigate the whole field	/

4. Model Design

● Framework and Logic

A well-designed sprinkler system can apply water uniformly to the soil surface and is capable of applying enough water to meet the peak demands of the crop without producing excess runoff. Generally, good-design considers such factors as pressure, nozzle size and spacing, the numbers of sprinklers, wind, air, temperature, water uses rates, and so on. Therefore, an efficient “hand move” sprinkler system is the result of good system design, proper irrigation scheduling, and careful operation and timely maintenance.

To provide guidelines on irrigation planning and management, a framework and logic chart is

developed as follows:

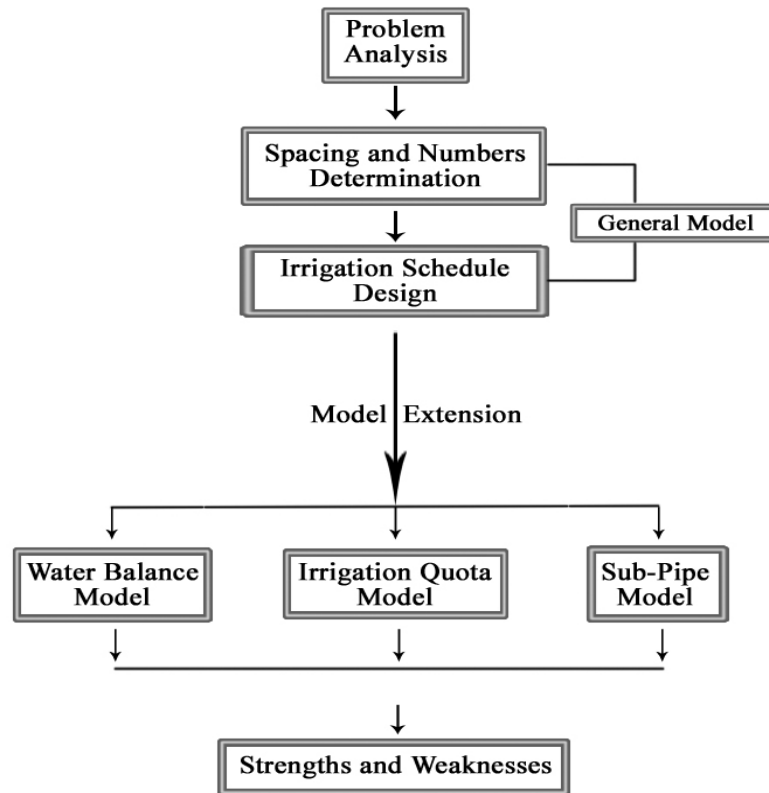


Figure 1: Framework and logic chart

As indicated in figure 1, the logic chart includes three primary components, which will be discussed in detail in our paper.

- **General Model**

- **Determine the number of sprinklers and spacing between sprinklers**

- **Maximizing Uniformity Coefficient**

We know that the spray drainage of a sprinkler is an approximate circle, and the amount of water decreases from its center to the edge. This causes the ununiformity of the spray area. Thus we need a parameter to measure the water's uniformity of the irrigated field.

First we define ρ_i as the Precipitation Rate (Pr) of sprinkler irrigation at the observation point i , which means the rate at which water is sprayed from the nozzle. Following the Precipitation Rate formula (Hill, R.W. 1991):

$$\rho_i = \frac{q \eta}{A} = \frac{q \eta}{\pi R^2} \quad (1)$$

where q is the flow rate of the nozzle, η represents the sprinkler Application Rate which varies from 70 percent to 90 percent, according to field conditions, A is the spray area, here we consider it as a circle, and R is the distance from the observation point to the center of the circle.

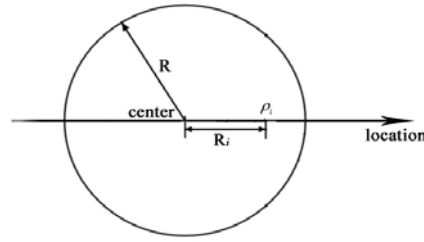


Figure 2: The spray area of one sprinkler

Then we use the “Uniformity Coefficient” named “Christiansen Coefficient” (Christiansen, 1942) to measure the applied water’s uniformity. It can be determined as follows:

$$C_u = 1 - \frac{\Delta\rho}{\bar{\rho}} \quad (2)$$

where C_u denotes the Uniformity Coefficient, $\bar{\rho}$ is the average Precipitation Rate (Pr) of the spray area, and $\Delta\rho$ represents the variance of the Precipitation Rate (Pr) in the spray area.

In order to improve the water’s uniformity of the field, we should seek for the maximum value of Uniformity Coefficient subject to some restrictions.

Plant growth in the field needs proper amount of water, in this problem, no part of the field should receive more than 0.75 cm per hour of water, and each part of the field should receive at least 2 cm of water every four days. These determine the upper and lower limit of the discharge of the nozzle.

Thus we develop a model to seek the maximum value of C_u .

$$\max C_u = 1 - \frac{\Delta\rho}{\bar{\rho}} \quad (3)$$

$$\text{S.T.} \quad \begin{cases} \rho_i = \frac{q\eta}{\pi R_i^2} + \frac{q\eta}{\pi R_{i-1}^2} \\ q = \frac{Q}{n} \\ i = 1, 2, 3, \dots, n \\ R_i + R_{i-1} = l \\ R_i \in (0, R) \\ R_{i-1} \in (0, R) \end{cases}$$

where Q is the total flow rate of the water source, n is the number of sprinklers. s denotes the spacing between sprinklers, R_i and R_{i-1} represent the distance from point ρ_i to the center of each circle respectively.

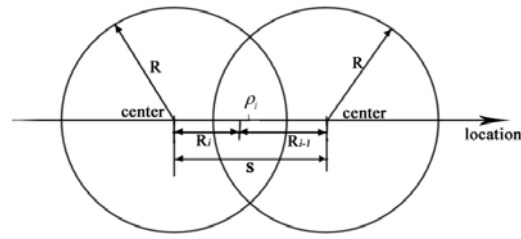


Figure 3: The spray of two sprinklers

➤ Prove and Conclusion

The number of the sprinklers and the spacing between them can be determined by using two-dimensional search, while it's difficult to realize through program. So we use an approximate method to simplify this solution.

The Uniformity Coefficient C_u is used to measure the uniformity of water that applied to the field, which is relative with the Precipitation Rate ρ_i . Following the formula of ρ_i , we give the actual Precipitation Rate trendline as below:

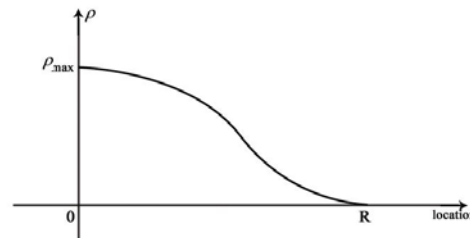


Figure 4: The Precipitation Rate trendline

Figure 4 shows the relationship between the Precipitation Rate and the distance from the nozzle to its edge. It's easy to understand that the farther the location is, the less water it can be irrigated.

To simplify the calculation, we substitute a beeline for the actual trendline:

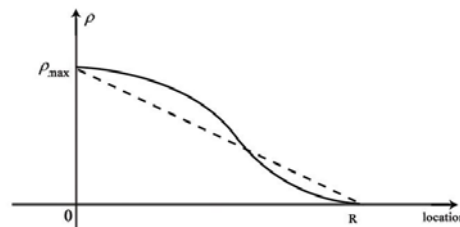


Figure 5: The simple Precipitation Rate trendline

Correspondingly, we denote a circle with the radius of R to represent the area that a sprinkler can cover. Taking two sprinklers for instance, let the two circles overlapped with each other, and the distance of two circles from the centers is the radius R :

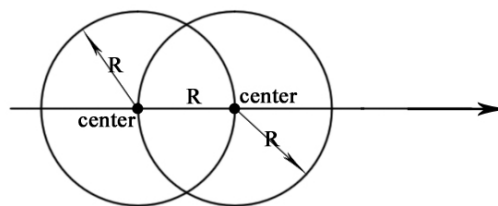


Figure 6: The spray area of two sprinklers with the spacing R

Combining the Figure 5 with Figure 6, we get:

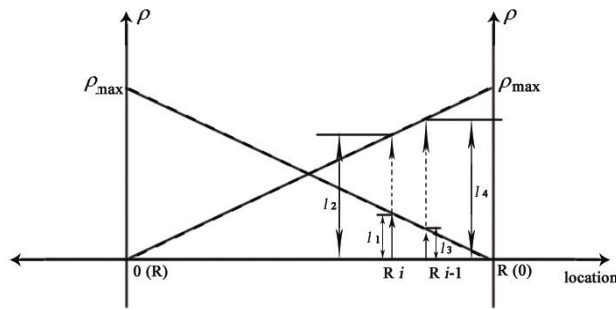


Figure 7: Precipitation Rate of the spray area of two sprinklers with the distance R

As indicated in Figure 7, the Precipitation Rate of the two observation points in R_i and R_{i-1} are:

$$\begin{cases} \rho_i = l_1 + l_2 \\ \rho_{i-1} = l_3 + l_4 \end{cases} \quad (4)$$

Apparently,

$$l_1 + l_2 = l_3 + l_4$$

i.e.

$$\rho_i = \rho_{i-1}$$

Therefore, we get the conclusion: the water can be applied in a high uniformity when the spacing between two adjacent nozzles equals to the spray radius. Generally, considering the influence of the wind, the spacing between the two sprinklers is $(0.9R, 1.1R)$ [Reference 7].

Furthermore, without considering the influence of wind, so the spacing between two adjacent sprinklers s can be determined as:

$$s = R \quad (5)$$

The numbers of sprinklers n :

$$n = \left\lfloor \frac{L}{s} \right\rfloor \quad (6)$$

where L denotes the total length of the pipes. In our model, we put all of the pipes together, which means $L = 20m$

■ Design optimal Irrigation Schedule

➤ “Rectangle Area”

It's crucial that the water should be applied as uniformly as possible. Because the volume of water will reduce linearly with the distance to the center increasing, which will cause an imbalance growth of the crop in different part of the field, thus it will lead to a yield reduction. For better understanding of the water spray situation, we draw the figure composed of six nozzles:

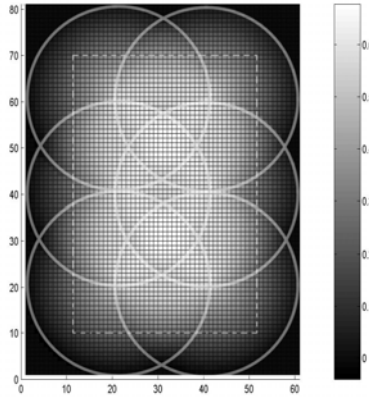


Figure 8: The uniformity of the area irrigated by six sprinklers

As indicated in Figure 8, based on our conclusion above, there exists an area in the center that denoted by dashed line, where the water can be applied uniformly as possible by the water's overlapping. It is what we expect.

Now, we determine the size of the “rectangle area”. As what we have done before, the number of the sprinklers is 5, and the spacing between sprinklers is 5 m . Also, the pipe is 20 meters long. Define a ring with a radius R as the area a sprinkler covers, and let the rings intersect with each other, whose distance between two centers is R . Following the agricultural criterion (Last Electronic Revision: June 1996), we get:

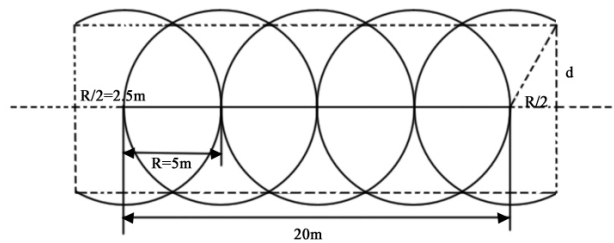


Figure 9: Sketch map of determining the size of the “rectangle area”

From Figure 9, we get:

$$d = \sqrt{R^2 - \left(\frac{R}{2}\right)^2} = \frac{\sqrt{3}}{2} R = \frac{5\sqrt{3}}{2} m$$

Therefore, we determine the size of the “rectangle area” as follows;

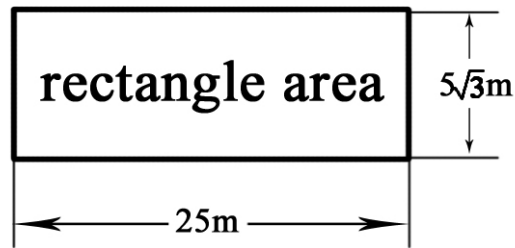


Figure 10: The size of the “rectangle area”

So we can use this area where the water can be applied uniformly to cover the whole field with the goal of minimizing the moving times.

➤ Principles of the Scheme

As mentioned in the problem, what we should do is that “find an irrigation schedule algorithm that minimizes the amount of time required by a rancher to maintain the irrigation system”, that is to say, minimizing the time that required is crucial in this problem. In general, the time for ranchers to maintain the irrigation system is composed of two parts:

- (1) The time of positioning the pipe, which requires us to manage a proper schedule and decide where to move the pipe;
- (2) The time spending on equipments, such as un-coupling, moving and re-coupling the pipe. Apparently, the less times we need to un-couple move and re-couple the pipe, the fewer time would be required during the whole irrigation.

According to the analysis above, we propose two principles for our schedule as follows:

(1) Principle of Optimal Positioning

This principle is aimed to minimize the time of moving the pipe. Suppose this situation: a “rectangle area” is placed in an arbitrary area of this field. We have to decide the next location where the “rectangle area” would be moved. The next location should be as close to the last location as possible, so as to reduce the time of moving. The comparison can be seen through the figure below:

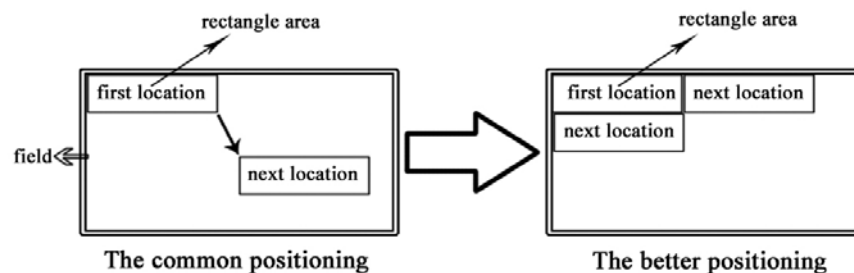


Figure 11:Two methods of positioning

(2) Principle of Minimum Times of Moving

This principle is proposed to minimize the times that set or move the pipes, which demands us to design a schedule of irrigating the field with minimum times. It means using the “rectangle area”

in Figure 10 above to cover the whole field as fewer times as possible;

➤ **Expatiation of the Scheme**

The disadvantage of the “hand move” system is that it requires a great deal of time and effort to move and set up the equipment at regular intervals. Therefore, we should propose a scheme to conquer it. At the same time, the uniformity of water that applied should be satisfied, which can be realized by the “rectangle area”. The process can be described as the figure below:

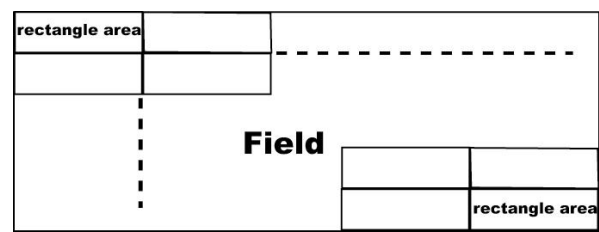


Figure 12:The process of “covering” the field by the “rectangle area”

So we summarize the aim of our scheme: minimize the amount of time required by the ranchers to maintain the irrigation system, demanding us to cover the given field by using the “rectangle area” as less times as possible. Considering the characteristic of rectangle, there are two positioning method to fill the given field: in the horizontal line (HL) and in the vertical line (VL), which are illuminated by figure 13:

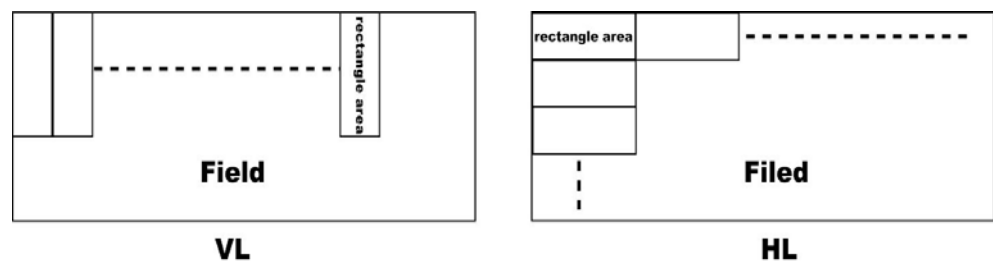


Figure 13:Two directions of positioning

Following our principles, we adopt “Heuristic Algorithm” to solve the problem. Take HL positioning method for instance, first of all, we give an initial position for the “rectangle area”. Then, we detect the next location: if the “rectangle area” can cover the next location, we continue the detecting process; if the next location is not large enough to be covered by the “rectangle area”, we define the next location as the “buffer area”. With a continuous iteration, most of the field can be filled with the “rectangle area”. Afterward, we need to deal with the “buffer area”. We use both HL and VL methods to cover the “buffer area” to increase Cover Rate a . In fact, in our scheme, if $a > 95\%$, we think it’s acceptable, which will be explained below. Likewise, the VL method can be disposed by a similar positioning method.

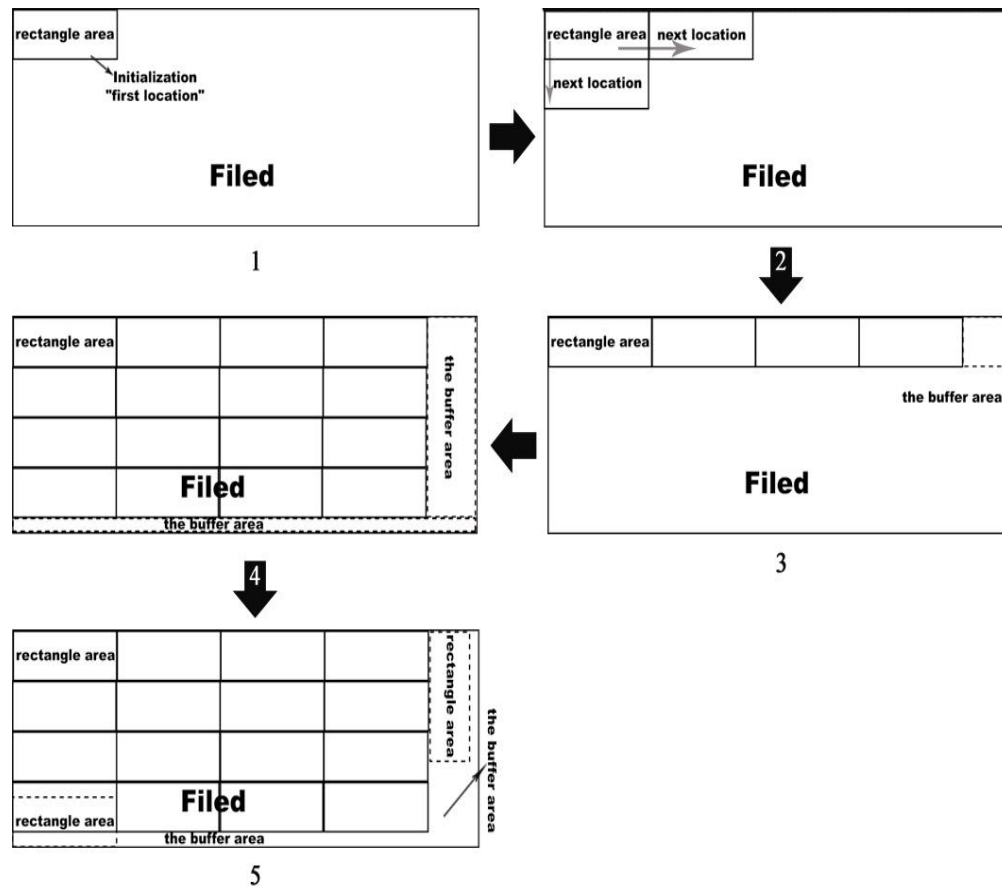


Figure 14:The process of "covering" the field

The HL algorithm is described as follows:

Step 1: Initialize the given filed, give an initial location to the "rectangle area";

Step 2: Detect the next location.

If the next location is large enough been covered by the "rectangle area", continue the detecting process; otherwise, define the next location as "buffer area", sign and register it.

Step 3: Keep detecting the next location until the field is divided completely by two parts: the total "rectangle area" and "buffer area".

Step 4: Judge if the Cover Rate a reached the acceptable Cover Rate.

If it has reached, go to Step 6; otherwise, go to Step 5.

Step 5: Compare the Overlap Rate b caused by covering different buffer areas, then choose the low Overlap Rate buffer area, and cover it with the "rectangle area", then go to Step 4.

Step 6: Output the total numbers of the "rectangle area" T , the Cover Rate a , the Overlap Rate b .

Figure 15 shows the flow chart:

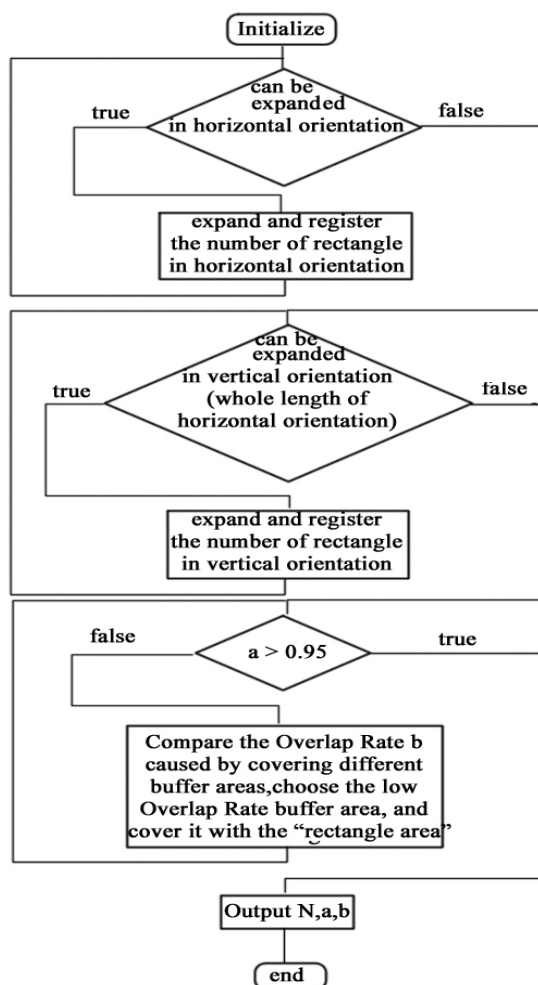


Figure 15: The flow chart of our algorithm

The original program code will be presented in the Appendix.

5. Model Solution

● Number and Spacing

■ Discussion about Spray Radius

Now, we discuss the possible values of the spray radius R . We find two types of sprinklers in RainBird homepage. The performance data table [Reference 3] is shown below:

Table 1:The performance data of the RainBird Handmove Sprinklers

Straight Bore Nozzle (SBN-3)

(Stream Height: 2.1m)

BARS @ Nozzle	NOZZLE SIZE											
	2.78 mm (7/64")			3.18 mm (1/8")			3.57 mm (9/64")			3.97 mm (5/32")		
	Rad. (m)	Flow (lps)	Flow (m ³ /h)	Rad. (m)	Flow (lps)	Flow (m ³ /h)	Rad. (m)	Flow (lps)	Flow (m ³ /h)	Rad. (m)	Flow (lps)	Flow (m ³ /h)
1.4	10.8	0.10	0.35	11.1	0.13	0.46	11.6	0.16	0.58	11.9	0.20	0.72
1.5	11.0	0.10	0.37	11.2	0.13	0.48	11.6	0.17	0.61	12.0	0.21	0.75
2.0	11.4	0.12	0.43	11.5	0.15	0.56	12.0	0.20	0.70	12.4	0.24	0.87
2.5	11.6	0.13	0.48	11.8	0.17	0.62	12.3	0.22	0.79	12.8	0.27	0.97
3.0	11.8	0.14	0.52	12.1	0.19	0.68	12.6	0.24	0.86	13.1	0.30	1.06
3.5	11.9	0.16	0.56	12.2	0.20	0.73	12.8	0.26	0.93	13.4	0.32	1.15
4.0	12.0	0.17	0.60	12.4	0.22	0.79	13.0	0.28	1.00	13.6	0.34	1.23
4.5	12.0	0.18	0.64	12.6	0.23	0.83	13.1	0.29	1.06	13.7	0.36	1.30
5.0	12.1	0.19	0.67	12.7	0.24	0.88	13.3	0.31	1.11	13.9	0.38	1.37
5.5	12.2	0.20	0.71	10.1	0.26	0.92	13.4	0.32	1.17	14.2	0.40	1.44

From the table, we know that when the inner diameter is 3.97mm , while the pressure is 4.0bars (about 400 Kilo-Pascal's), $R = 13.6\text{m}$. R may be longer when inner diameter is 6 mm . Considering that the field is $80\text{ meters by }30\text{ meters}$, and the total length of the pipe is 20 m , 13.6 m seems too long. So we recommend that the spray range of the nozzle $R = 5\text{m}$.

■ Numbers and Spacing

Following the Equations (5) and (6), the spacing between two adjacent sprinklers s can be determined as:

$$s = R$$

The numbers of sprinklers n :

$$n = \left\lfloor \frac{L}{s} \right\rfloor$$

Substitute $L = 20\text{m}$, and $R = 5\text{m}$, we obtain:

$$s = 5\text{m}, n = 5$$

We draw the figure of the Precipitation Rate (Pr) to see the uniformity of the spray irrigation. It is shown as follows:

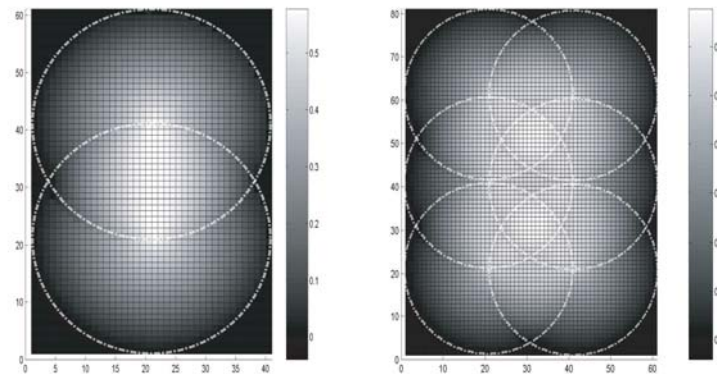


Figure 16: The uniformity of the area that irrigated by different number of Sprinklers

The left part of Figure 16 shows the spray condition of two sprinklers, every single circle expresses the spray area of a sprinkler (denoted by dash-dot-line in the figure above). The Precipitation Rate (Pr) in overlapped area between two circles is nearly the same as that of the area near the center of the circle, which means the uniformity in these areas is quite high.

The right part of Figure 16 is composed of six sprinklers. Similarly, in the center part of the figure, the Precipitation Rate (Pr) is almost the same and the spray area (denoted by dash-dot-line the figure above) near the center is an approximate rectangle.

● Irrigation Schedule Design

■ Estimate of Irrigation Times

The Precipitation Rate (Pr) is used to measure the water requirement. It (Pr) is the rate at which

water is sprayed from the nozzle, averaged as inches per hour, over the area covered by one nozzle. It is important to consider the Precipitation Rate when designing a sprinkler system, since water will run off if applied faster than the soil can absorb it. Precipitation rate can be calculated by using the Precipitation Rate formula (Hill, R.W. 1991.):

$$\text{Pr (inches/hr)} = 96.3 \times \text{nozzle flow rate (gpm)/area covered (ft}^2\text{)}$$

where 96.3 is a constant that converts gallons per minute to inches per hour

This equation also can be represented as:

$$\rho = \frac{q \eta}{A} = \frac{q \eta}{\pi R^2} \quad (7)$$

where

ρ : The Precipitation Rate (Pr), m/h ;

q : Nozzle flow rate, m^3/h ;

η : Application Rate, application efficiency varies with field conditions, 80 percent is a reasonable value;

R : The radius of the spray area of the nozzle, m . It has been estimated in our paper above.

The radius of the spray area, as we estimated above, is about 5 meters. Thus for a single sprinkler, the area it can cover is $A = \pi R^2 = \pi \times 5^2 = 78.5 m^2$, then the nozzle flow rate can be confirmed through the equation below:

$$q = \frac{Q}{n} = \frac{150}{5} \times \frac{1}{1000} \times 60 m^3 / h = 1.8 m^3 / h$$

where n is the number of sprinklers, $\frac{1}{1000}$ and 60 are constants that convert mm^3 to m^3 .

Therefore, the Precipitation Rate can be calculated following Equation (7):

$$\rho = \frac{q \eta}{A} = \frac{q \eta}{\pi R^2} = \frac{1.8 \times 0.8}{78.5} m / h \approx 0.0183 m / h = 1.83 cm / h$$

Then the times of irrigation can be determined as:

$$t = \frac{q_i}{1.83}$$

where t is the time duration each time, q_i is the irrigation amount per time.

For the sake of time uniformity, the field should be irrigated by stages so that each part of the field can be irrigated rotatively, which will require us to determine the times of irrigation more scientifically, also the Precipitation Rate (Pr) should be between the upper and lower limits. In this problem, the crop should receive no more than $0.75 cm/h$, and receive at least $2 cm$ of every 4 days. Thus, we determine the irrigation amount for each time at the average level, which is $0.5 cm$ per time, and the irrigation duration

$$t = \frac{0.5}{1.83} = 0.273 \text{ hours}$$

■ Irrigation Schedule

➤ Irrigation Schedule

We get the minimal times that move or set up the “hand move” system. The result table is as follows:

Table 2: Result of the problem

$R = 5m$					
HL			VL		
T_m	a	b	T_m	a	b
13	98.95%	18.31%	12	96.81%	11.43%

where T_m represents the times of positioning the “rectangle area”, a denotes the Cover Rate and b is the Overlap Rate of this field.

First of all, we explain why the Cover Rate is acceptable when $a > 95\%$. In our scheme, we don’t consider the pervasion of the water, which exists in actual situation. So, the actual “rectangle area” may be larger, that is to say, the actual Cover Rate a may be higher. In another word, if $a > 95\%$, we think the field is well covered by the “rectangle area”. Therefore, both of this two positioning method can achieve the same goal.

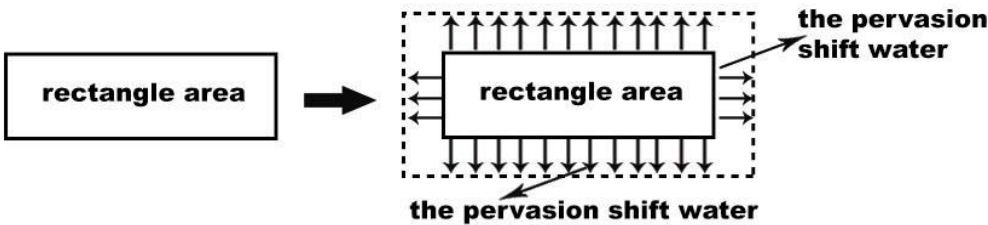


Figure 17:Process of water pervasion

From Table 2, we know that: If we adopt the HL positioning method, the rancher needs to move the pipe 12 times, while the times is 13 if the VL method is adopted; the Cover Rate in VL positioning method equals to 18.31%, which is greater than that of VL positioning method’s Overlap Rate 11.43%.

Therefore, we propose the rancher applying HL positioning method to irrigate the field. Following the principles, the field is divided into 3 parts: A, B, and C.

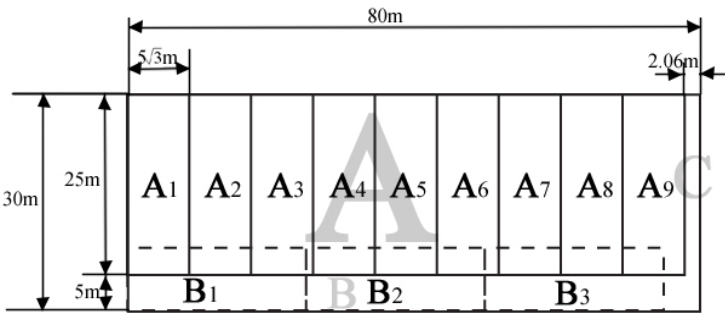


Figure 18:Three parts of the field

Based on Figure 18, we give the irrigation schedule as follows:

Table 3: Detailed schedule for the irrigation system

Location		Moment t												
		t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}	t_{12}	t_{13}
A	A_1	Pipe												
	A_2		Pipe											
	A_3			Pipe										
	A_4				Pipe									
	A_5					Pipe								
	A_6						Pipe							
	A_7							Pipe						
	A_8								Pipe					
	A_9									Pipe				
B	B_1										Pipe			
	B_2											Pipe		
	B_3												11 minutes later	Pipe

Where

- ✓ The schedule aims at the irrigation timing for one day. Following the schedule, the rancher can irrigate the whole field per time in a minimal time required;
- ✓ “Pipe” denotes that the sprinkler system irrigates the area at the corresponding moment;
- ✓ The regular intervals equal to

$$t_2 - t_1 = t_3 - t_2 = \cdots = t_i - t_{i-1} = t_{12} - t_{11} = 0.273hours ;$$

- ✓ As indicated in Figure 18, B_3 overlaps with A_7 , A_8 and A_9 , if B_3 is irrigated in 49 minutes, this area may receive water more than 0.75 *cm* per hour. Therefore, we recommend that the rancher should wait for 11 minutes later, that is to say, the extra waiting time $T_w = t_{13} - t_{12} = 0.183hours$.

➤ **Amount of Time**

Based on schedule, the total time required to irrigate the whole field can be calculated as:

$$T = T_m \times t + T_w = 12 \times 0.273 + 0.183 = 3.459 \text{ (hours)}$$

6. Model Extensions

In our General Model, some of the parameters, such as the Precipitation Rate and Irrigation Period, are determined by estimating. In order to calculate these important parameters for design a irrigation schedule more accurately, we propose three other models: By considering climate condition and soil-water relationship, Water Balance Model is employed to calculate the water requirements per day, which will be more scientific to supply water during the whole growth season; By considering different crops and soil texture, Irrigation Quota Model is proposed to determine the Irrigation Period for one whole season; By dividing the field into small sections, Sub-Pipes Model is designed in order to reduce water requirement and cut down the expense of the irrigation system.

● **Water Balance Model:**

Good design considers such factors as pressure; nozzle size and spacing; wind, air temperature and

humidity (day versus night); soil intake rate; crop rooting depth and water use rates. The design pertains to the climate–soil–plant system and can be applied to deduce the amount and timing of irrigation needed to ensure adequate crop growth.

In our General Model above, we don't take such factor as air condition, temperature, soil-water-relationship and water use rates into account. We just simply estimate the average irrigation requirement. While the schedule needs to be adjusted when these factors vary each day. Thus we want to develop our model by considering these factors and design an advanced model: Water Balance Model, to determine the irrigation requirement for each day. Apparently, the requirement is more scientific and proper for the crop to be irrigated. We need the monthly mean and standard deviation weather data include the following:

- Reference crop evapotranspiration (mm/day);
- Average daily air temperature (°C);
- Total precipitation in each month (mm);
- Number of rainy days in each month (days).

In Water Balance Model, we consider the influence of climate such as air temperature, rain falling and evaporation, then design an advanced model and establish a long-term irrigation requirement for the field. By taking these factors into account, we can determine the water balance in soil more scientifically, and supply water more accurately. These would improve water use rates; that is to say, we can have the water applied more effectively so as to economize the amount of water.

We can use daily weather data to simulate the water balance of this field. The daily simulation procedure includes three programming loops, which are the command areas within the simulated irrigation project, the crops within each command area, and the days from planting to harvest for each crop. The output from this model includes relative crop yield and crop irrigation water requirements. Notably, both are the required inputs for the following optimization model.

When estimating irrigation water requirements, the on-farm daily soil water balance may be maintained by applying Equation (8):

$$SMD_j = SMD_{j-1} - IRR_j - PE_j + ET_j + EWS_j \quad (8)$$

where SMD_j and SMD_{j-1} denote the soil moisture depletion values at the j th and $(j-1)$ th days, IRR_j represents the depth of irrigation water at the j th day. PE_j denotes the effective rainfall at the j th day. ET_j is the evapotranspiration rate at the j th day. And finally, EWS_j is the evaporation rate of wet soil surface after irrigation and/or rainfall on the j th day.

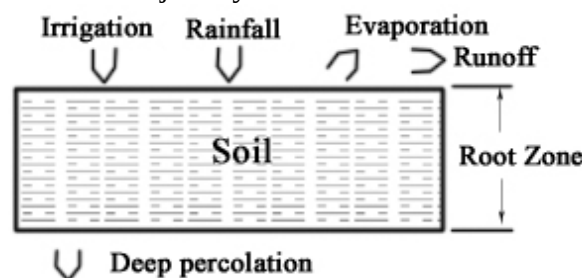


Figure 19: Schematic representation of the root zone water-balance components

Based on weather data of, Hargreaves' equation (Hargreaves et al., 1985) is employed to calculate the reference crop coefficient. The equation is presented as:

$$ET_0 = 0.0023R_a(T + 17.8)\sqrt{T_{\max} - T_{\min}} \quad (9)$$

where ET_0 denotes the reference crop coefficient. R_a represents the extraterrestrial radiation (equivalent mm/day). T is the main daily air temperature ($^{\circ}\text{C}$). $T_{\max}$ and $T_{\min}$ denote the maximum and minimum daily air temperature ($^{\circ}\text{C}$) respectively.

Based on the daily simulation procedure, the daily basal crop coefficient K_{cb} represents the effects of the crop canopy on evapotranspiration and varies with the seasons. Here we don't consider any canopy that can have effect on crops, so $K_{cb}=0$.

To calculate both the potential crop evapotranspiration (ET_c) and actual crop evapotranspiration (ET_{ca}), the daily reference crop evapotranspiration is employed as presented in Equation (10) and (11), respectively:

$$ET_c = (K_{cb} + K_s) \times ET_0 \quad (10)$$

$$ET_{ca} = (K_{cb} \times K_a + K_s) \times ET_0 \quad (11)$$

where ET_c denotes the potential crop evapotranspiration (mm/day), K_{cb} represents the basal crop coefficient (here, $K_{cb}=0$), and ET_0 is the reference crop evapotranspiration (mm/day), and

$$K_a = \frac{\ln(100 \times \frac{\theta_j - \theta_{wp}}{\theta_{fc} - \theta_{wp}} + 1)}{\ln(101)} \quad 0 < K_a \leq 1 \quad (12)$$

$$K_s = (1 - K_c) \times \left\{ 1 - \left(\frac{t_w}{t_d} \right)^{1/2} \right\} \times F_w \quad (13)$$

where ET_{ca} denotes the actual crop evapotranspiration, K_a represents the soil moisture stress coefficient and K_s is the coefficient for evaporation rate from a wet soil surface after irrigation and/or rainfall. Furthermore, θ_j denotes the soil moisture by volume at the j th day, θ_{fc} and θ_{wp} represent the soil moistures by volume at the field capacity and wilting point, and t_w is the time in days since wetting, due to irrigation and/or rainfall. Finally, t_d denotes the time in days required for the soil surface to dry after irrigation and/or rainfall.

Regarding on-demand irrigation scheduling, irrigation should be performed when soil moisture depletion (SMD) exceeds the allowable depletion (AD) initially. The amount that required, or application depth, in a given irrigation (IRR), and allowable depletion (AD), are described mathematically by Equations (14) and (15), respectively:

$$IRR_j = \frac{SMD_j}{(E_c \times E_a)} \quad (14)$$

$$AD_j = (\theta_{fc} - \theta_{wp}) \times RZ_j \times MAD_{stage} \quad (15)$$

where IRR_j denotes the irrigation requirement on the j th day, SMD_j represents the soil moisture depletion on the j th day, E_c is the conveyance coefficient of soil and E_a denotes the water application efficiency. RZ_j represents the crop's root depth on the j th day and MAD_{stage} is the maximum allowable soil water depletion within each stage.

The cumulative irrigation water requirement for the command field is the sum of the irrigation requirements for each day, which can be represented as:

$$IRR_{total} = \sum_{j=1}^n IRR_j \quad (16)$$

● Irrigation Quota Model

In our General Model, we estimate the irrigation period is four days, which is not accurate enough to calculate the irrigation water requirement and may cause unnecessary waste. And also, the irrigation period varies throughout the season, because the water requirement in every growth period for a crop is different. In order to supply water more properly, and to minimize the irrigation waste, we introduce "Irrigation Quota" to determine the water requirement.

Irrigation Quota is the total water requirement of the crop growth for one season. In another word, it is the sum of net irrigation each time. It can reflect the water requirement of a crop and the performance of an irrigation plan. Scientific Irrigation Quota can improve the yield of crops and reduce soil moisture depletion caused by over-irrigation. Consequently Irrigation Period and Irrigation Timing can be determined via Irrigation Quota.

■ Determine Irrigation Quota

Irrigation Quota can be calculated via equation as:

$$m = rh(p_1 - p_2) \times \frac{1}{\eta} \quad (17)$$

where m is Irrigation Quota (m^3 / ac) of the field, r denotes the density of the soil, p_1 and p_2 represent the lower and upper limit of the soil water holding capacity, η is the sprinkler application efficiency, which varies from 70 percent to 90 percent, according to field condition. 80 percent is a reasonable value.

■ Determine Irrigation Timing

Irrigation Timing mainly concerns by some factors such as climate of the area, soil moisture and the growth condition of crops. We'll discuss the Irrigation Timing from these aspects:

(1) Texture of soil

The texture of soil to be irrigated is very important in determining when to irrigate. Different types of soil have different capacity to hold water: the water in sand-soil has fast percolation and evaporation, while that in clayed soil is slow. This difference will cause different water condition in root zone and require us to establish different Irrigation Timing.

Table 4 lists abilities of different soil types to store water. (W. Trimmer and H. Hanson, 1994)

Table 4: Soil water storage capacities*

Soil		Readily available moisture/ft.
General description	Texture class	
Light, Sandy	Coarse Sand	0.7 inches
	Fine Sand	0.9
Medium, Loamy	Fine Sandy Loam	1.5
	Silt Loam	2.0
Heavy, Clay	Clay Loam	2.3
	Clay	2.0
	Peats and Mucks	2.5

*Values given are for deep, uniform soil profiles. Layering and changes in soil texture within the profile may increase or decrease effective available water.

The crop root zone determines soil depth from which the crop can draw moisture. Table 5 shows the root zones that mature crops depend on for 90 percent of their water needs. Early in the season, annual crops have shadow root zones and approach the values in Table 5 only when they reach full cover. (W. Trimmer and H. Hanson, 1994)

Table 5:Root zone depths for selected crops

Crop	Root zone* (ft)	Time to reach mature root zone	Allowable depletion %
Alfalfa	4.0	0	60
Beans	2.5	50 days after planting	50
Corn	3.0	10 days after tasseling	50
Grapes	3.0	0	65
Orchard	6.0	0	50–65
Potatoes	2.0	80 days after planting	30–40
Pasture/ Turf	2.0	0	60
Small Grains	3.0	heading	50
Sugar Beets	3.0	10 days after planting	50

*The root zone can be limited by shallow soils, compaction layers, and dry soil—all of which reduce amount of water available to the crop, thus requiring more frequent irrigations.

Plants will show signs of wilting and drought stress before they use all available water stored in the soil. Table 5 shows percent of total available moisture that different crops can withdraw without suffering yield reduction.

(2) Soil Moisture

Soil moisture should be measured initially and monitored regularly to determine the available soil moisture. Soil moisture blocks, neutron probes, tension meters, or the feel method with soil probing will all work. Some methods work better than others with different soil types. If the soil humidity reduces to the lower limit of the proper soil moisture, irrigation is needed then.

(3) Growth Condition of Crops

Every crop has its own critical period and fastigium in water requirement. Generally, wheat requires irrigation most during trefoil stage to heading stage, which is the best irrigation timing for this kind of crop. While the best irrigation timing for corn is from interodeelongation stage to grain filling stage, and for legume it is from branching stage to blooming stage.

■ Determine Irrigation Period

Irrigation Period represents the interval between two irrigations. It can be estimated as:

$$T = \frac{m}{w} \eta \quad (18)$$

where m is the Irrigation Quota, which can be calculated by Equation (17), w denotes the water requirement of crop per day, and η is the sprinkler Application Rate, which varies from 70 percent to 90 percent according to field condition. 80 percent is a reasonable value.

After estimating these parameters, we can design a more scientific schedule for irrigation, both reducing the water waste and improving the yield of crop.

● Sub-Pipe Model

In our General Model, we consider that the pipe is 20 meters long and cannot be disconnected into small parts. The time required by a rancher to maintain the system is calculated only through considering the times of un-coupling and re-coupling the pipe. In fact, the time of moving pipe into different area of the field varies with the distance, which also influence the total time required to maintain the system. In this section, we'll take its influence into account by employing our algorithm designed in the General Model into every section of the field.

For the time of moving pipe taken into account, we should prevent moving pipe for so long distance. Thus the pipe can be considered been disconnected into several parts which can be called sub-pipes, and the field which need to be irrigated can be partitioned into several sections. For a single sub-pipe, we restrict it irrigating a single section of the field, which will cut the distance of move the pipe and reduce the total time of moving. We design the Sub-Pipe Model and present its location of as follows:

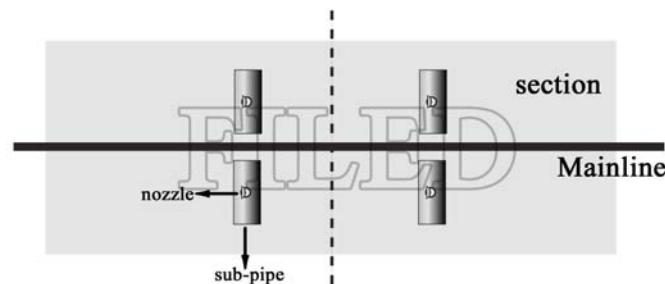


Figure 20: Sub-Pipe Model design

This design of sub-pipes also has other strengths:

- The mainline locates in the center of the field, which will reduce the whole length of the pipe resulting in the reduction of the head loss. Those will cut down the expense of the irrigation system.
- The field is partitioned into several sections, we employ our algorithm in every section to search the minimum times of un-coupling and re-coupling sub-pipes, this strategy will reduce the total time required and let the water applied more uniform.

7. Strengths and Weaknesses

Model Strengths

- Our “Heuristic Algorithm” can deal with any field, no matter what it shape or square is. It has flexibility.
- By using Our General model, the field can be irrigated with a high uniformity, and economize the water requirement.
- Our irrigation schedule is easy to operate by one rancher.
- The Water Balance Model have a full consideration of Soil-Water relationships, irrigation timing and amount depends on soil water holding capacity, weather, and crop growth progress; calculate the total Irrigation Requirement and the yield reduction per season.
- The Sub-Pipe Model reduces the total time required of maintaining the irrigation system and the whole length of the pipe, cut down the expense of this system.

Model Weaknesses

- Our model doesn't take some factors such as terrain, water loss during transport into account.
- We determine the spray range simply without considering the wind's influence.

8. References

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Appendix

Appendix 1: HL Program

%this program is aimed to find T,a,b

```
R=5; %the radius of the sprinkler spray area
per=0.95; %the minimum percent covered with water
width=R*sqrt(3); %width of the rectangle area
length=5*R; %length of the rectangle area
width_n=fix(30/width);
length_n=fix(80/length);
N=width_n*length_n;
buffer_width=30-width*width_n;
buffer_length=80-length*length_n;
per_cover=(width*length*N)/2400; %the percent area covered has been irrigated
per_dou=0;
n_1=0;
k=0;
for i=1:20
    if per_cover>=per %judge per_cover if it has reached the min percent covered with water
        fprintf('You should move the pipe %d times.\n',N);
        fprintf('%f percent has been irrigated.\n',per_cover*100);
        fprintf('%f percent has been irrigated more than one time.\n',per_dou*100);
        break
    else
```

```

if buffer_length>width
    n_1_1=fix(30/length);
    n_1_2=fix(buffer_length/width);
    n_1=n_1_1*n_1_2;
    N=N+n_1;
    per_cover=per_cover+n_1*width*length/2400;
    buffer_length=buffer_length-n_1_2*width;
else
    if n_1>0
        if k==0;
            mi=min([length*(width-buffer_width) length*(width-buffer_length) width*(length-(30-
n_1_1*length))]);
            if mi==length*(width-buffer_width)
                n_2=length_n;
                N=N+n_2;
                per_cover=per_cover+n_2*width*length/2400-width_n*length*(width-
buffer_width)/2400;
                per_dou=per_dou+width_n*length*(width-buffer_width)/2400;
                buffer_width=0;
            else
                if mi==length*(width-buffer_length)
                    n_3=n_1_1;
                    N=N+n_3;
                    per_cover=per_cover+n_3*width*length/2400-n_3*length*(width-
buffer_length)/2400;
                    per_dou=per_dou+n_3*length*(width-buffer_length)/2400;
                    buffer_length=0;
                else
                    n_4=n_1_2;
                    N=N+n_4;
                    per_cover=per_cover+n_4*width*length/2400-n_4*width*(length-(30-
n_1_1*length))/2400;
                    per_dou=per_dou+n_4*width*(length-(30-n_1_1*length))/2400;
                    k=1;
                end
            end
        else
            mi=min([length*(width-buffer_width) length*(width-buffer_length)]);
            if mi==length*(width-buffer_width)
                n_2=length_n;
                N=N+n_2;
                per_cover=per_cover+n_2*width*length/2400-width_n*length*(width-
buffer_width)/2400;

```

```

        per_dou=per_dou+width_n*length*(width-buffer_width)/2400;
        buffer_width=0;
    else
        mi==length*(width-buffer_length)
        n_3=n_1_1;
        N=N+n_3;
        per_cover=per_cover+n_3*width*length/2400-n_3*length*(width-buffer_length)/2400;
        per_dou=per_dou+n_3*length*(width-buffer_length)/2400;
        buffer_length=0;
    end
end
else
    mi=min([length*(width-buffer_width) length*(width-buffer_length)]);
    if mi==length*(width-buffer_width)
        n_2=length_n;
        N=N+n_2;
        per_cover=per_cover+n_2*width*length/2400-n_2*length*(width-buffer_width)/2400;
        per_dou=per_dou+n_2*length*(width-buffer_width)/2400;
        buffer_width=0;
    else
        n_3=fix(30/length);
        N=N+n_3;
        per_cover=per_cover+n_3*width*length/2400-n_3*length*(width-buffer_length)/2400;
        per_dou=per_dou+n_3*length*(width-buffer_length)/2400;
        buffer_length=0;
    end
end
end
end
end
end
end

```

Result:

When $R=5m$

You should move the pipe 13 times.

98.958333 percent has been irrigated.

18.315940 percent has been irrigated more than one time.

Appendix 2: VL Program

%this program are aimed to find T,a,b

R=5; %the radius of the sprinkler spray area

per=0.95; %the minimum percent covered with water

width=R*sqrt(3); %width

length=5*R; %length

```

width_n=fix(80/width);
length_n=fix(30/length);
N=width_n*length_n;
buffer_width=80-width*width_n;
buffer_length=30-length*length_n;
per_cover=(width*length*N)/2400; %the percent area coverd has been irrigated
per_dou=0;
n_1=0;
k=0;
for i=1:20
    if per_cover>=per        %judge per_cover if it has reached the min percent coverd with water
        fprintf('You should move the pipe %d times.\n',N);
        fprintf('%f percent has been irrigated.\n',per_cover*100);
        fprintf('%f percent has been irrigated more than one time.\n',per_dou*100);
        break
    else
        if buffer_length>width
            n_1_1=fix(80/length);
            n_1_2=fix(buffer_length/width);
            n_1=n_1_1*n_1_2;
            N=N+n_1;
            per_cover=per_cover+n_1*width*length/2400;
            buffer_length=buffer_length-n_1_2*width;
        else
            if n_1>0
                if k==0;
                    mi=min([length*(width-buffer_width)  length*(width-buffer_length)  width*(length-(80-
n_1_1*length))]);
                    if mi==length*(width-buffer_width)
                        n_2=length_n;
                        N=N+n_2;
                        per_cover=per_cover+n_2*width*length/2400-width_n*length*(width-
buffer_width)/2400;
                        per_dou=per_dou+width_n*length*(width-buffer_width)/2400;
                        buffer_width=0;
                    else
                        if mi==length*(width-buffer_length)
                            n_3=n_1_1;
                            N=N+n_3;
                            per_cover=per_cover+n_3*width*length/2400-n_3*length*(width-
buffer_length)/2400;
                            per_dou=per_dou+n_3*length*(width-buffer_length)/2400;
                            buffer_length=0;

```

```

else
    n_4=n_1_2;
    N=N+n_4;
    per_cover=per_cover+n_4*width*length/2400-n_4*width*(length-(80-
n_1_1*length))/2400;
    per_dou=per_dou+n_4*width*(length-(80-n_1_1*length))/2400;
    k=1;
end
end
else
    mi=min([length*(width-buffer_width) length*(width-buffer_length)]);
    if mi==length*(width-buffer_width)
        n_2=length_n;
        N=N+n_2;
        per_cover=per_cover+n_2*width*length/2400-width_n*length*(width-
buffer_width)/2400;
        per_dou=per_dou+width_n*length*(width-buffer_width)/2400;
        buffer_width=0;
    else
        mi==length*(width-buffer_length)
        n_3=n_1_1;
        N=N+n_3;
        per_cover=per_cover+n_3*width*length/2400-n_3*length*(width-buffer_length)/2400;
        per_dou=per_dou+n_3*length*(width-buffer_length)/2400;
        buffer_length=0;
    end
end
else
    mi=min([length*(width-buffer_width) length*(width-buffer_length)]);
    if mi==length*(width-buffer_width)
        n_2=length_n;
        N=N+n_2;
        per_cover=per_cover+n_2*width*length/2400-length_n*length*(width-buffer_width)/2400;
        per_dou=per_dou+width_n*length*(width-buffer_width)/2400;
        buffer_width=0;
    else
        n_3=fix(80/length);
        N=N+n_3;
        per_cover=per_cover+n_3*width*length/2400-n_3*length*(width-buffer_length)/2400;
        per_dou=per_dou+n_3*length*(width-buffer_length)/2400;
        buffer_length=0;
    end
end
end

```

```
    end  
  end  
end
```

Result:

When $R=5\text{ m}$

You should move the pipe 12 times.

96.814882 percent has been irrigated.

11.438294 percent has been irrigated more than one time.